Housing: Rent or Buy — Cheat Sheet

Financial Algebra · Unit 7: Housing: Rent or Buy

Essential vocabulary

Lease (rental)
A contract to rent a home for a set term at a set rent, with rules for both tenant and landlord

Security deposit
Money paid to a landlord up front to cover damage or unpaid rent, generally refundable

Renters insurance
Insurance that covers a tenant's belongings and liability, not the building

Mortgage
A loan used to buy real estate, secured by the property

Principal and interest (P&I)
The portion of a mortgage payment that repays the loan and pays interest

PITI
Principal, interest, property taxes, and homeowners insurance — the full monthly housing payment

Down payment
The part of a home's price paid in cash at purchase

Private mortgage insurance (PMI)
Insurance protecting the lender, typically required when the down payment is under 20% on a conventional loan

Closing costs
Fees paid to complete a home purchase, often 2–5% of the price

Property tax
A local tax on real estate based on its assessed value

Equity
The part of a home's value that the owner truly owns: market value minus loan balance

Appreciation
An increase in the value of an asset over time

Opportunity cost (housing)

Key formulas

Max housing payment	0.28 × gross monthly income
DTI	$\frac{\text{all monthly debt payments}}{\text{gross monthly income}}$
Mortgage P&I	$\text{PMT} = \frac{L \cdot i}{1 - (1 + i)^{(-n)}}$
PITI	$\text{P\&I} + \frac{\text{annual tax}}{12} + \frac{\text{annual insurance}}{12}$ (+ PMI, HOA)
Equity	value − loan balance
Opportunity cost	cash × ((1 + r)^t − 1)
Net cost of buying	outflows − equity at sale + opportunity cost

Typical cost ranges (illustrative)

Closing costs	2–5% of price
Selling costs	5–7% of sale price
Maintenance	1–2% of value per year
PMI	≈ 0.5–1% of loan per year if < 20% down

The return given up by tying cash up in a down payment and closing costs instead of investing it

Front-end ratio
Housing payment divided by gross monthly income; many lenders prefer about 28% or less

Debt-to-income ratio (DTI)
All monthly debt payments, including housing, divided by gross monthly income

HOA fee
A monthly fee paid to a homeowners association for shared maintenance and amenities

Maintenance reserve
Money set aside for home repairs; a common rule of thumb is 1–2% of home value per year

Common misconceptions

- Renting is always a waste.
- Homes always appreciate.
- Mortgage payment = cost of owning.
- Approval = affordability.

Worked example

1. $350,000 home, 20% down → $280,000 loan at 6.75%, 30 yr
2. P&I ≈ $1,816.07
3. Tax 1.2% → $350/mo; insurance $1,500 → $125/mo
4. PITI ≈ $2,291.07; + maintenance ≈ $291.67 → ≈ $2,582.74

Questions to ask before deciding

- How long will I stay?
- What is the full monthly cost (PITI + maintenance)?
- How much cash do I need to close and still keep an emergency fund?
- What could the down payment earn elsewhere?
- What if prices fall?

Decision tips

- Get preapproved and compare lenders.
- Keep an emergency fund after closing.
- Budget maintenance from day one.
- Run optimistic and pessimistic scenarios.

VALUES AND RATES ARE ILLUSTRATIVE. ALL EXAMPLES ARE FICTIONAL. MR. K'S CLASSROOM · FINANCIAL ALGEBRA